KVL Experiment:

To verify Kirchhoff's voltage law (KVL) using a series electrical circuit.

component	value
V_1	12 V
R_1	2 KΩ
R_2	4 KΩ
R_3	6 KΩ
Ammeter	1 mA
voltage across R_1	2 V
voltage across R_2	4 V
voltage across R_3	6 V

Check by calculations:

Total resistance:

$$R_T = 2K + 4K + 6K = 12K\Omega$$

current:

$$I = \frac{12}{12K} = 1mA$$

voltage drops:

$$V_1 = 1mA \times 2K\Omega = 2V$$

$$V_2 = 1mA \times 4K\Omega = 4V$$

$$V_3 = 1mA \times 6K\Omega = 6V$$

Now check KVL :

$$V_1 + V_2 + V_3 = 2 + 4 + 6$$

$$= 12V$$

source voltage = 12V

so :

$$12V = 2V + 4V + 6V \qquad V = V_1 + V_2 + V_3$$

$$12V = 12V \quad //$$

voltage across R_1

$$V_1 = IR_1$$

$$V_1 = 1mA \times 2K\Omega$$

$$V_1 = 2V$$

voltage across R_2

$$V_2 = IR_2$$

$$V_2 = 1mA \times 4k\Omega$$

$$V_2 = 4V$$

voltage across R_3

$$V_3 = IR_3$$

$$V_3 = 1mA \times 6K\Omega$$

$$V_3 = 6V$$

Result :

$$12V = 2V + 4V + 6V$$

conclusion :

The sum of the voltage drops across the resistors is equal to the

applied source voltage. Therefore, Kirchhoff's voltage law is verified successfully.